Dattatray Suresh Mongad,

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Current academic position

2017-Present	Ph.D. (Microbiology)	National Centre for Cell Science, Pune, India.

Education

2015-2017	M.Sc. (Bioinformatics)	Bioinformatics Centre, SPPU, Pune, India.
2012-2015	B.Sc. (Biotechnology)	Fergusson College, Pune, India.

Skills

Bioinformatics skills

Genomics, Microbiome (Amplicon & Metagenome), Transcriptomics, Metabolome, Gene-expression and SNPs analysis, Machine Learning

Programming/soft skills

R (Shiny apps), Python, HTML5, PHP, Perl
Operating system: Linux and Windows

Research interests

I am a highly motivated researcher interested in developing tools/workflows to automate analysis and visualiation of microbiome data. With Microbiology/Biotechnology and Bioinformatics experience, I have gained good experience in designing workflows for the analysis of microbiological omics data. We have developed tools for functional prediction of microbiome data (MicFunPred) and probiotic prediction using machine learning (ProBioPred). I have gained experience in analyzing amplicon, metagenome, transcriptome sequence data and metabolome data. I have also trained myself in data analysis using R & Python and software development using Python (OOP). I am looking forward to skilling myself in statistics and machine learning.

Research experience

2017-Present	Ph.D. (Microbiology) Topic: Development of a tool for	National Centre for Cell Science, Pune, India.

functional prediction of microbiome data.

Projects

Bachelor's thesis

2014-2015 Biodiversity studies of alkaliphilic bacteria from Lonar lake.

Master's thesis

2016-2017 ASTEP: A step towards Allele Specific T-cell Epitope Prediction using Support Vector Machines.

Ph.D. thesis

2017-Present MicFunPred: A conserved approach to predict functional profiles from 16S rRNA gene sequence data.

Student supervision

2018	Co-supervision	Ms. Simran Cheema, Dr. D. Y. Patil Biotechnology & Bioinformatics Institute, Pune, India. Project title: Effect of different maternal microbial communities on the development of infants oral and gut microbiota. Present position: Marie-Curie ESR, Queen's University Belfast, Northern Ireland.
2019	Co-supervision	Ms. Pooja Dangare, Bioinformatics Centre, SPPU, Pune. Project tile: Genome annotation of a Human gut isolate from Indian origin.

Professional activities

Planning committee member: Workshop on "Statistical methods in microbiome research".
Instructors:
1. Prof. Susan Holmes, Statistics, Stanford University, California.
2. Prof. Leo Lahti, University of Turku, Finland.

Publications

Journal articles

1. Chaudhari, Diptaraj S., Dhiraj P. Dhotre, Dhiraj M. Agarwal, Akshay H. Gaike, Devika Bhalerao, Parmeshwar Jadhav, **Dattatray Mongad** et al. "Gut, oral and skin microbiome of Indian patrilineal families reveal perceptible association with age." Scientific reports 10, no. 1 (2020): 1-13.
2. Kumbhare, Shreyas V., Dhrati V. Patangia, **Dattatray S. Mongad**, Abhijeet Bora, Ashish R. Bavdekar, and Yogesh S. Shouche. "Gut microbial diversity during pregnancy and early infancy: an exploratory study in the Indian population." FEMS Microbiology Letters 367, no. 3 (2020): fnaa022.

Book chapters

1. "Computational Techniques Used for Microbial Diversity Analysis." Microbiome-Host Interactions, by **Dattatray S. Mongad** et al., 1st ed., CRC Press, 2021, pp. 21–35.

Working papers

1. **Dattatray S. Mongad**, Nikeeta S. Chavan, Nitin P. Narwade, Dhiraj P. Dhotre and Yogesh S. Shouche (2021). MicFunPred: A conserved approach to predict functional profiles from 16S rRNA gene sequence data.
2. **Dattatray S Mongad**; Shaima A Rifaie; Shrikant Bhute; Shailesh S Mantri; Yogesh S Shouche; William B Whitman; Kamlesh Jangid (2021). K-shuff reveals heterogeneous soil bacterial communities along the slopes of an analog for Martian impact craters
3. Diptaraj Chaudhari, Dhiraj Dhotre, Dhiraj Agarwal, Akshay Gaike, Devika Bhalerao, Parmeshwar Jadhav, **Dattatray Mongad**, Himangi Lubree, Vilas Sinkar, Ulhas Patil, Sundeep Salvi, Ashish Bavdekar, Sanjay Juvekar, and Yogesh Shouche (2021). Gut, oral and skin microbiome of Indian patrilineal families reveal perceptible association with age.
4. Khemraj; Rishi Praveen; Shukla Geeta; Shivaprakash M.R.; **Mongad Dattatray**; Kaur Amrita (2021). Deciphering transcript isoform switching and its usage during biofilm lifestyle of Candida glabrata through a global transcriptomic approach.
5. Runa Antony, **Dattatray Mongad**, Aritri Sanyal, Dhiraj Dhotre and Meloth Thamban (2021). Holed up, but thriving: Impact of multitrophic cryoconite communities on glacier elemental cycles.